Subversion: Version Control Rethought

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Greg Stein, CollabNet, Inc

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Agenda

- What is Subversion?
- Overview
- Basic Setup
- Simple Examples
- Comparison to CVS
- Better than CVS
- Additional Tools
- Wrap up

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What is Subversion?

- Version control system
- Working model similar to CVS
- Intended to replace CVS
- But more than just a replacement!

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Why should it be like CVS?

- CVS is very popular
- Conceptually simple
- Easy to use for the simple case
- Demonstrated capability [of the model] over wide-area networks

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Overview

- Open Source
  - many reasons...
- http://subversion.tigris.org/
- Designed for remote users
- Unreserved checkouts
  - Similar to CVS
  - checkout, edit, merge, commit
- Command-line client included
- "Third-party" GUIs in development
- Apache-based network server

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Basic Setup

- Install Apache 2.0.x
- Install Subversion
- Create a repository: `svnadmin create /path/to/repository`
- Configure Apache (next slide)
- *Have fun...*

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Configuring Apache

- Pretty simple configuration:

```
LoadModule dav_svn_module modules/mod_dav_svn.so
<Location /svn/repos>
  DAV svn
  SVNPath /path/to/repository
</Location>
```

- See `INSTALL` for further options

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Example

Check out a working copy, edit a file, then commit:

```
$ svn checkout http://svn.example.com/repos/
project
$ cd project
$ jed brain.txt
$ svn commit -m "new plan to conquer the world"
```


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Subversion vs CVS

- Most CVS features
- Directory versioning
- Metadata
- Atomic commits
- Branching and tagging are cheap
- Designed for the network
- Better binary file handling
- Layered library design

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Feature Comparison

- Feels familiar to CVS users
- Missed one feature: `cvs annotate`
- Omitted reserved checkout features (`edit`, `watch`) and the `history` subcommand
- Option handling has been changed and simplified
- More subcommands
- Some features/tasks are handled differently

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Additional Subcommands

- `copy`, `move`
- `merge`
- `resolve`
- `mkdir`
- `propset`, `propget`, `proplist`, `propdel`, `propedit`
- `revert`
- `switch`
- `info`

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Difference: Revision Numbers

- Global revision number, rather than per-file
- Allows you to talk about "revision 6387"
- Unique identifier for a state of your project
  - Simple way to tag your project (next slide)
- Each revision corresponds to a single commit
  - A "change set"
  - Associated properties: author, log message, date

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Difference: Tagging and Branching

- Subversion implements "cheap copies"
- Branches are just copies of the main trunk
  - Make changes in the copy
  - Merge the changes back to the main trunk
- Tags are simply copies which are never changed

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Sample Repository Layout

```
http://svn.example.com/repos/project/
   trunk/
      source/
      docs/
      buildtools/
   branches/
      issue-1003/
      gstein/
   tags/
      alpha-1/
      1.0.0/
      1.0.1/
```

The `trunk`, `branches`, `tags` layout is the recommended form, but you are free to structure the repository in any way that suits you best.

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Difference: Authentication

- CVS uses a custom authentication mechanism
  - Part of CVS's custom (non-standard) protocol
  - "I LOVE YOU" or "I HATE YOU"
  - pserver sends passwords in the clear
- Alternate authentication schemes: ksever, gserver, SSH tunneling
- Subversion uses HTTP as its primary network protocol
  - Integrates with existing authentication systems
  - Standardized!
- SSH tunneling is also available for Subversion

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Difference: Modules

- CVS Modules
  - Live in `CVSROOT/modules`
  - Only the administrator can edit the modules
  - Only used at checkout time
- Subversion "externals"
  - Directory property: `svn:externals`
  - Users can define, edit, delete, and inspect them
  - Attach to any directory
  - Versioned, as with any other property
  - Changes are detected during `svn update`

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Difference: Keywords

- CVS keywords are automatically expanded
  - Users must explicitly disable this behavior
  - Risk of destroying a binary file
  - Unversioned flag
- Subversion keywords are optionally expanded
  - User must proactively enable the expansion
  - User can specify which keywords to expand
  - Controlled by a versioned property: `svn:keywords`

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Additional Tools

- cvs2svn
- ViewCVS -- now does Subversion, too
- Hook scripts: commit emails, simple ACLs, reserved checkouts, repository backup
- Libraries, scripting, and svnlook
- GUI clients such as RapidSVN, TortoiseSVN, JSVN, SVNUP, Subclipse, AnkhSVN, etc

Future

- Reserved checkouts
- Better merge tracking
- More complete i18n/l10n support
- Robust ACL support
- Complete WebDAV/DeltaV compatibility and support
- Relational database for the backend

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Thanks! Q&A...

Questions?

Remember: http://subversion.tigris.org/

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